

1. Introduction

The Satellite Learning Laboratory is a satellite engineering ecosystem that supports workforce development, training and hands on experimentation. It also provides a tool to support the development of spacecraft subsystems, hardware, software and operations procedures.

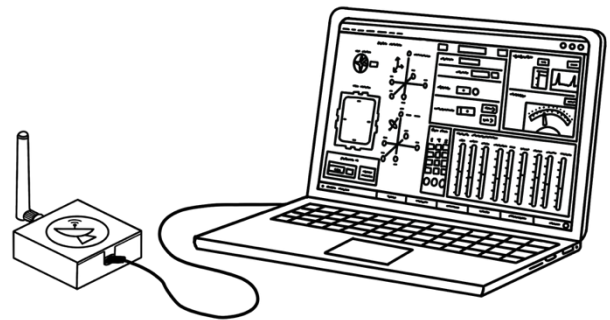
2. Aims and objectives

This lab will familiarise the user on how to configure the ground station to the same frequency as the demonstration Satellite Simulator.

3. Equipment needed

The following equipment will be required for this lab:

- Ground Station Simulator
- SCOTTI Satellite Operations Software



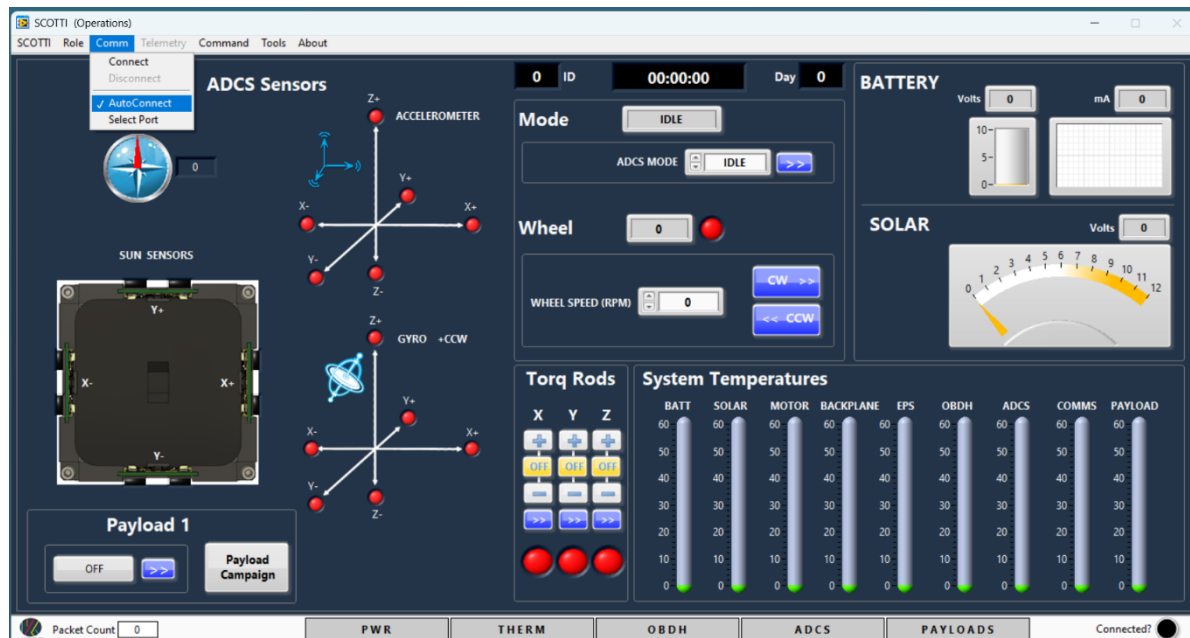
4. Instructions

Using the Quick Start Guide, chapter 6, perform the following steps:

1. Locate the Ground Station Radio.
2. Plug the unit into the computer that has the ground station software installed.
3. Start the computer, locate the SCOTTI_SATLL application and run it.
4. SCOTTI will now load.
5. SCOTTI will ask if you wish to "Load Default Database?", Select "Yes".
6. The SCOTTI is now running and shows the SCOTTI (Operations) summary page.

At this point we are not connected to the ground station. We need to connect to the ground station. However, the communications is by default it is in "auto connect" mode. We need to turn this off before proceeding.

1. Locate the "Comm" menu on the top left of the SCOTTI screen, uncheck "Auto Connect" option.



2. Now in the same menu click "Select Port".
3. Select the "COM" port that the radio is plugged into. If the Ground Station is the only thing plugged into the USB port only the one option may exist.

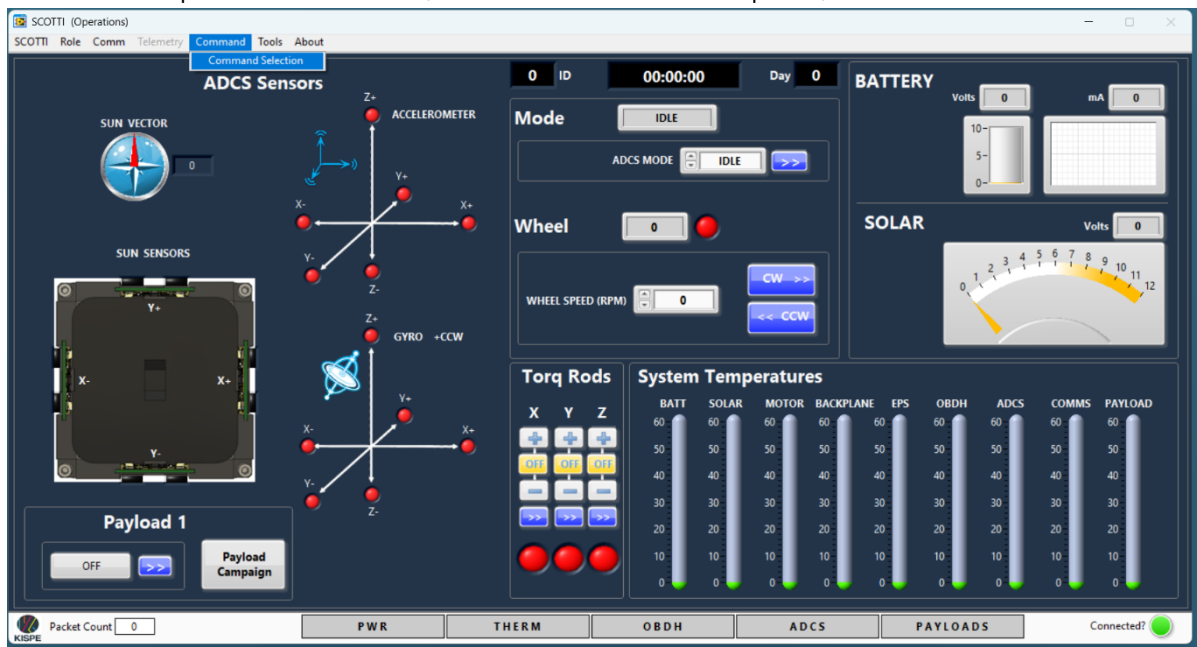
If there are multiple – look at the ports without the ground station plugged in and then with the ground station plugged in, the new port is the ground station.



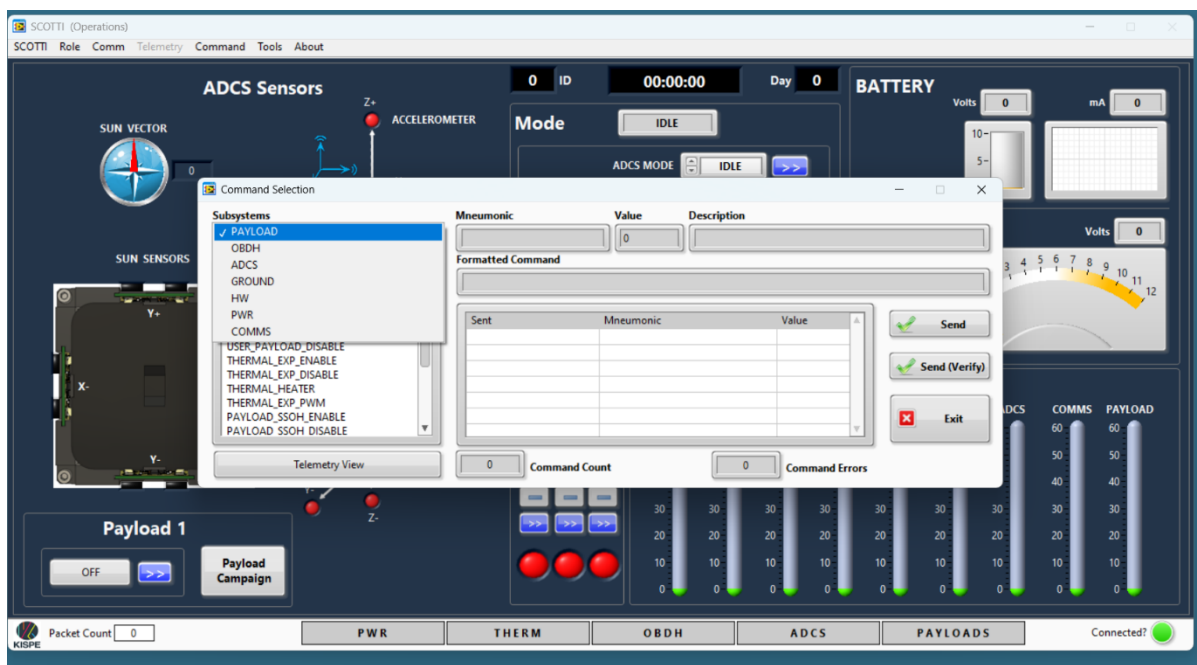
4. Within the "Comm" menu, select "Connect".
5. The light on the ground station box should now be green. You will not be receiving telemetry though.

We are now connected to the ground station, but not the simulator. Now we want to change the Ground Station frequency to match the demonstration satellite simulator.

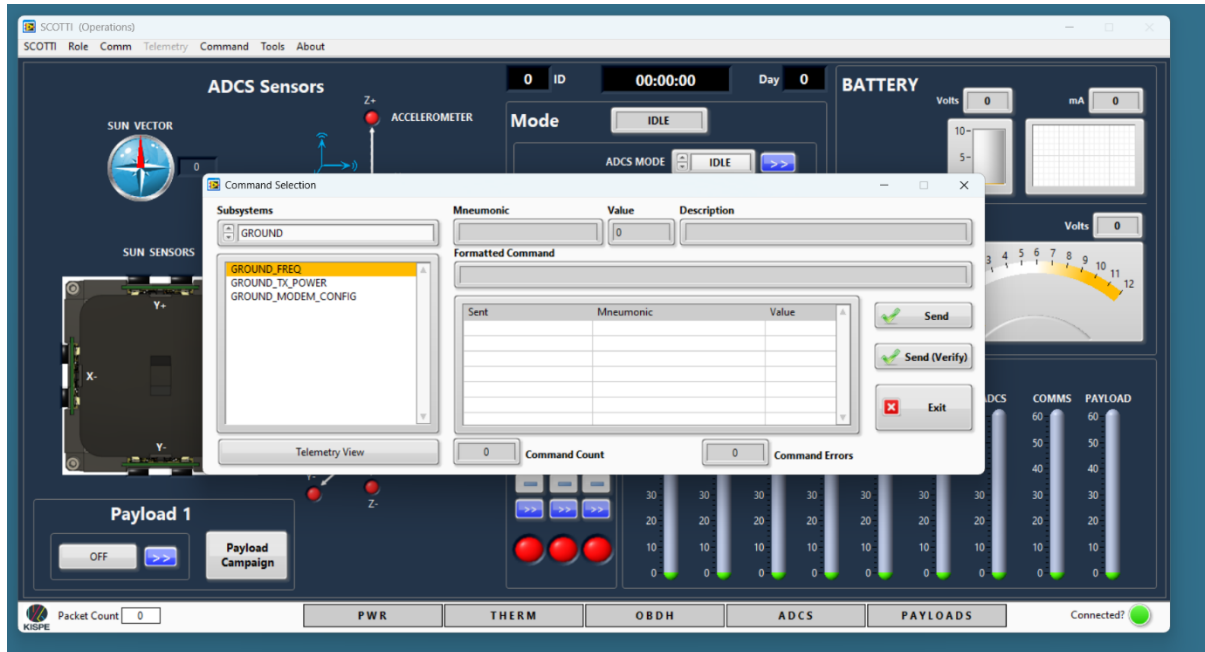
6. From the top menu of SCOTTI, select the Command Option, then Command Selection.



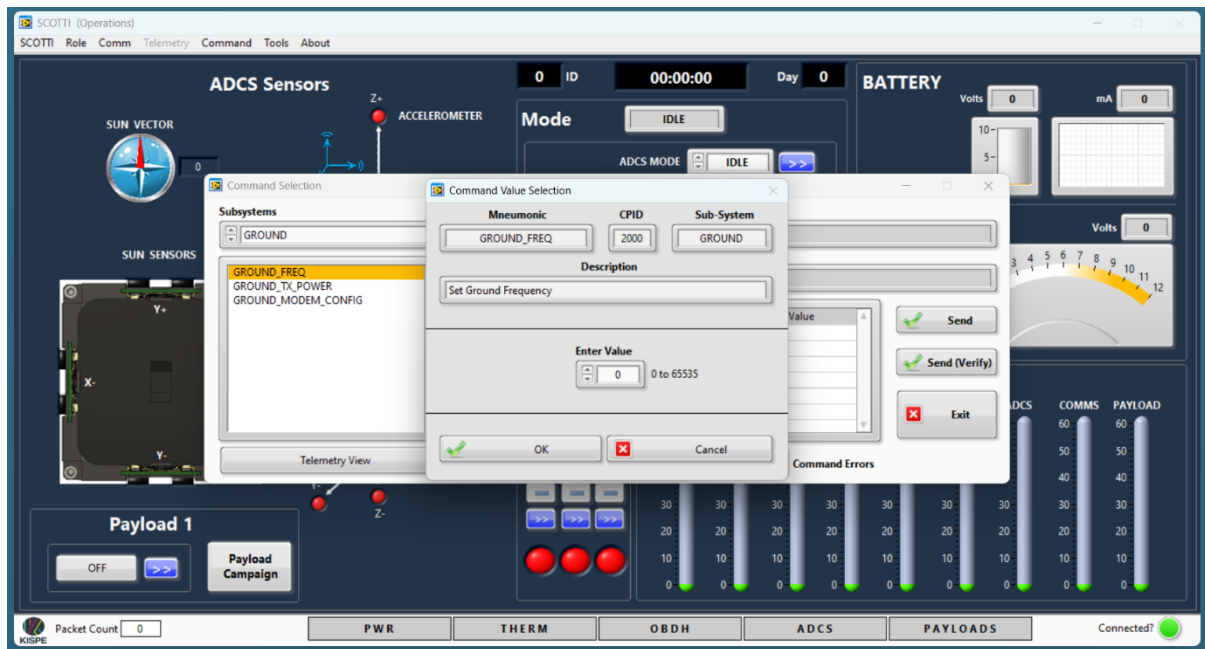
7. The command selection screen then comes up. In the top left sub system box, select "GROUND".



8. Once the GROUND commands are selected the following page comes up:



9. Select the "Ground_FREQ" option. Double Click for the command window to come up.



10. The Instructor will tell you what value to enter into the "Enter Value" box. Enter this value and then click OK.

Frequency number calculation
 $0 = 864 \text{ MHz}$ and increments in 1 kHz steps.
 Therefore to get 866.25 MHz enter 2250

11. Then select the "Send" command.

12. The Ground station indicator light should go from green to purple.

At this point you are not connected to a Satellite Simulator, so you will not be collecting telemetry.

Now we need to connect to the demonstrator's Satellite Simulator.

13. Within the "Comm" menu, select "Connect"

14. The light on the ground station box should now be flashing purple.

15. The SCOTTI (Operations) Screen should now be alive and telemetry will be displayed on the screen.

16. The "Connected" indicator in the bottom right hand corner of the window will be green, signifying that data is being received.

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SATELLITE LEARNING LABORATORY